Predavanja

ustni izpit: 40%

predstaviteev: 20% (20 minut po 3 osebe)

Vaje

Projekt 40%

# Logic in computer science

- Correctness of software/hardware/systems
  (watch Xavier Leroy: In search of sofware prfection)
- Developing proof asistants

Propositional logic (natural deduction proof system)
Simple type theory : Curry-Howard correspondance
(propositions as types)

# Propositional logic

propositional formulas $\phi, \psi, \theta$

propositional variables $p$

atomic formula:

$$\phi ::= p \mid \psi \wedge \theta \mid \psi \vee \theta \mid \psi \to \theta \mid \neg\theta \mid \bot$$

TAVTOLOGIJA - validna formula (vedno res)

Natural deduction: proof system. Using formal rules we prove a formula $\phi$

We can prove formula $\Leftrightarrow$ it is a tavtology

Primer: Proof of $p \wedge q \to p \vee r$

1. Assume $p \wedge q$ true (assumption)
2. $p$ ($\wedge$elimination$_1$)
3. $p \vee r$ ($\vee$introduction$_1$)
4. $p \wedge q \to p \vee r$ ($\to$ introduction)

# Za konstruktivno logiko (Brez LEM

$\wedge$

$$\frac{\phi \quad \psi}{\phi \wedge \psi}\wedge i \qquad \frac{\phi \wedge \psi}{\phi}\wedge e_1 \qquad \frac{\phi \wedge \psi}{\psi}\wedge e_2$$

premises

conclusion

$\vee$

$$\frac{\phi}{\phi \vee \psi}\vee i_1 \qquad \frac{\psi}{\phi \vee \psi}\vee i_2 \qquad \frac{\phi \vee \psi \quad \boxed{\begin{matrix}\phi\\ \vdots\\ \theta\end{matrix}} \quad \boxed{\begin{matrix}\psi\\ \vdots\\ \theta\end{matrix}}}{\theta}\vee e$$

$\rightarrow$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \psi\end{matrix}}}{\phi \rightarrow \psi}$$

box discharges the assumption $\phi$

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi}\rightarrow e$$

modus ponens

$\neg$

$\neg\phi \equiv \phi \rightarrow \bot$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \bot\end{matrix}}}{\neg\phi}\neg i \qquad \frac{\neg\phi \quad \phi}{\bot}\neg e$$

$\bot$

$$\frac{\bot}{\phi}\bot e$$

NO FORMUA IN THE BOX gets used outside the box.

Primer $p \wedge (q \vee r) \rightarrow (p \wedge q) \vee (p \wedge r)$

1. $p \wedge (q \vee r)$ assumption
2. $p$
3. $q \vee r$
4. $q$ assumption
5. $p \wedge q$
6. $(p \wedge q) \vee (p \wedge r)$
7. $r$ assumption
8. $p \wedge r$
9. $(p \wedge q) \vee (p \wedge r)$ $\vee e$ (3) (4-6) (7-9)
10. $(p \wedge q) \vee (p \wedge r)$ $\rightarrow i$ (1-10)

## In classical logic

$$\frac{\boxed{\begin{matrix}\neg\phi\\ \vdots\\ \bot\end{matrix}}}{\phi}raa$$

Reductio ad absurdum

DN: prove LEM with raa

$$\frac{}{\phi \vee \neg\phi} \qquad \frac{\dfrac{\dfrac{\dfrac{}{\neg\phi}}{\bot}}{\phi}}{\phi \vee \neg\phi} \qquad \frac{\dfrac{}{\neg\phi}}{\phi \vee \neg\phi}$$

$$\phi \vee \neg\phi$$

A proof of formula $\phi$ from assumptions $\Gamma$ is given by sequence of formulas following the rules such that all open assumptions in the proof are from $\Gamma$

notation: $\vdash \phi$

Theorem: $\vdash \phi$ (including raa) $\overset{\text{soundnes}}{\underset{\text{completness}}{\Longleftrightarrow}}$ $\phi$ is tautology

$\Leftrightarrow$ $\phi$ is true under any assigment of truth validnes to propositional variables that make all formulas in $\Gamma$ true

Formulas $\phi ::= p \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \to \phi \mid \bot$

Types $A ::= \alpha \mid A \times B \mid A + B \mid A \to B \mid$

$$t : A$$

term type

$$\frac{s : A \quad t : B}{(s, t) : A \times B} \qquad \frac{t : A \times B}{pr_1(t) : A}\times e_1 \qquad \frac{t : A \times B}{pr_2(t) : B}$$

variable

$$\frac{\begin{array}{c} x : A \\ \vdots \\ t : B \end{array}}{\lambda x : A . t : A \to B} \to i \qquad \frac{t : A \to B \quad u : A}{t u : B} \to e$$

term

$$\frac{t : A}{in_1(t) : A + B} \qquad \frac{t : B}{in_2(t) : A + B}$$

$$\frac{s : A + B \quad \boxed{\begin{array}{c} x : A \\ \vdots \\ t : C \end{array}} \quad \boxed{\begin{array}{c} y : B \\ \vdots \\ u : C \end{array}}}{\text{case } s \, (in_1(x : A) \mapsto t \mid in_2(y : B) \mapsto u) : C}$$

Primer:

A derived term of type $A \times B \to B \times A$

| | | | |
|---|---|---|---|
| 1. | $x : A \times B$ | assumption | |
| 2. | $pr_1(x) : A$ | $\times e_1$ | (1) |
| 3. | $pr_2(x) : B$ | $\times e_2$ | (1) |
| 4. | $(pr_2(x), pr_1(x)) : B \times A$ | $\times i$ | (3) (2) |
| 5. | $\lambda x : A \times B . (pr_2(x), pr_1(x)) : A \times B \to B \times A$ | $\to i$ | (1-4) |

tega ne potrebujemo

For every type we have introduction rules, eliminatio rules and computation rules

computation rule for rewrite rule (longer arrow)

$\times$ : $pr_1(s,t) \longrightarrow s$

$pr_2(s,t) \longrightarrow t$

$\beta$-reduction

substitute $u$ for $x$ in $t$

$\rightarrow$ : $(\lambda x{:}A.t)u \longrightarrow t[x:=u]$

(before doing this first rename bound variable in $t$ that are distinct from variables in $u$)

$+$ : $\text{case}\ (in_1(s))\ (in_1(x) \mapsto t \mid in_2(y) \mapsto u) \longrightarrow t[x:=s]$

$0$ : $\dfrac{t:0}{\text{empty}(t):A}$

DM: Find canonical term of type

$A \times (B+C) \longrightarrow A\times B + A \times C$

Propositional logic ~ Simple Type theory

Predicate logic ~ Dependent type theory
(first order logic

Propositional logic formulas

$$\phi ::= p \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \to \psi \mid \bot$$

arity $\geq 0$

Predicate logic extends prop. logic with <u>predicates</u> expressing relations, and <u>constants and function</u> symbols expressing elements

arity $\geq 0$

Terms express elements

$$t ::= x \mid c \mid f(t_1, \ldots, t_n)$$

constant

function symbol of arity $n$

Formulas express propositions

$$\phi ::= P(t_1, \ldots, t_n) \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \to \psi \mid \bot \mid \forall x \phi \mid \exists x. \phi$$

Example: a language for expressing property of integers in predicate logic

Constants: $0, 1$

function symbols $+$, $\cdot$ of arity 2

example terms $z$, $x+y$, $x\cdot(y+(z+1))$

predicates $=$, $\leq$

example formulas $x \leq y$, $\exists y \exists z \exists w \exists v.\ x = yy + z\cdot z + w\cdot w + v\cdot v$

free variable

We could also define $\leq$ as

$$x \leq y \Leftrightarrow \exists x. \exists y. \exists v. \exists u.\ y = x + z\cdot z + w\cdot w + v\cdot v + u\cdot u$$

(vsa naravna števila sa vsote štirih kvadratov)

$x+1=0 \quad (x=-1)$

We will interpreate predicate logic in type theory

Prop ... type of propositions

$$\frac{\phi : Prop \quad \psi : Prop}{\phi \wedge \psi : Prop} \wedge \text{form} \qquad \frac{\phi : Prop \quad \psi : Prop}{\phi \vee \psi : Prop} \vee \text{form.}$$

$$\frac{\phi : Prop \quad \psi : Prop}{\phi \to \psi : Prop} \to \text{formation} \qquad \frac{}{\bot : Prop}$$

(1) $p : Prop$ assumption

(2) $q : Prop$ assumption

} open assumption are called a context

(3) $p \wedge q : Prop$ $\wedge$ formation (1) (2)

(4) $q \wedge p : Prop$ $\wedge$ formation (2) (1)

(5) $p \wedge q \to q \wedge p : Prop$ $\to$ formation (3) (4)

$$p : Prop, q : Prop \vdash p \wedge q \to q \wedge p : Prop$$

For predicate logic we also neen quantifiers

# Predicate logic

$$\frac{\boxed{\begin{array}{c} x:A \\ \Phi:Prop \end{array}}}{\forall x:A.\,\Phi:Prop}\ \forall\text{-formation} \qquad \frac{\boxed{\begin{array}{c} x:A \\ \Phi:Prop \end{array}}}{\exists x:A.\,\Phi:Prop}\ \exists\text{-formation}$$

(formation, not introduction

| | | |
|---|---|---|
| ① | $p: A \to Prop$ | assumption |
| ② | $x:A$ | assumption |
| ③ | $p\,x : Prop$ | $\to e$ |
| ④ | $\forall x:A.\,px : Prop$ | $\forall$-formation (2) (3) |

$$p: A \to Prop \vdash \forall x:A.\,px : Prop$$

(Insted of formation rules, we can include conectives and quantifiers as typed constants

$\to, \vee, \wedge : Prop \to prop \to Prop$

$\exists, \forall : (A \to Prop) \to Prop$ )

iz Primera: $\forall(x:A.\,p(x))$

Example proof of

$$p : A \to Prop,\ q : A \to Prop \vdash (\forall x : A.\, px) \to (\forall x : A.\, qx) \to (\forall x : A.\, px \wedge qx)$$

**3 judgments forms**

$t : A$

$\Phi : Prop$ "$\Phi$ is prop."

$\Phi$ "$\Phi$ is true"

| | | |
|---|---|---|
| 1 | $p : A \to Prop$ | assump. |
| 2 | $q : A \to Prop$ | assump. |
| 3 | $\forall x : A.\, px$ | assump |
| 4 | $\forall y : A.\, qy$ | assump |
| 5 | $z : A$ | assum |
| 6 | $p\,z$ | $\forall E$ (3) |
| 7 | $q\,z$ | $\forall E$ (4) |
| 8 | $p\,z \wedge q\,z$ | $\wedge I$ (6) (7) |
| 9 | $\forall z : A.\, pz \wedge qz$ | $\forall I$ (5)-(8) |
| 10 | $\forall y : A.\, qy \to \forall z : A.\, pz \wedge qz$ | $\to$ (4)-(9) |
| 11 | $\forall x : A.\, px \to (\ldots) \to (\ldots)$ | $\to$ (3)-(10) |

z se ne sme pojaviti v kontekstu preg

$$\frac{\boxed{\begin{array}{c} z : A \\ \vdots \\ \Phi \end{array}}}{\forall z : A.\, \Phi}\ \forall I$$

$$\frac{\forall x : A.\, \Phi \qquad t : A}{\Phi[x := t]}\ \forall E$$